> *restart*

An example: Computing rational recursion from C^2-finite equations

> *with*(*CodeTools*, *CPUTime*) *#to compute the CPU time of computations*

$$[CPUTime] \tag{1}$$

> *with*(*NLDE*, *DalgSeq*) *#the DalgSeq subpackge from NLDE*

$$[DalgSeq] \tag{2}$$

>

The C^2-finite equation.
The inputs for the linear algebra approach and the Groebner bases approach differ by the presence of the index variable in the C-finite coefficient.

> $C2feq1 := c[1](n)\cdot s(n+1) + c[0](n)\cdot s(n) = 0$

$$C2feq1 := c_1(n)\, s(n+1) + c_0(n)\, s(n) = 0 \tag{3}$$

> $C2feq2 := c[1]\cdot s(n+1) + c[0]\cdot s(n) = 0$

$$C2feq2 := c_1\, s(n+1) + c_0\, s(n) = 0 \tag{4}$$

The C-finite equations

> $Cfeq1 := c[1](n+2) = c[1](n+1) + c[1](n)$

$$Cfeq1 := c_1(n+2) = c_1(n+1) + c_1(n) \tag{5}$$

> $Cfeq2 := c[0](n+2) = c[0](n+1) + 3\cdot c[0](n)$

$$Cfeq2 := c_0(n+2) = c_0(n+1) + 3\, c_0(n) \tag{6}$$

>

The Groebner bases approach

> *CPUTime*(*DalgSeq*:-*CCfiniteToDalg*(*C2feq2*, $s(n)$, [*Cfeq1*, *Cfeq2*], [$c[1](n)$, $c[0](n)$]))

$$100.781,\ s(n+4)\, s(n+3)\, s(n+1)\, s(n) + s(n+4)\, s(n+2)^2\, s(n) - 4\, s(n+3)^2\, s(n+2)\, s(n) - 6\, s(n+4)\, s(n+2)\, s(n+1)^2 + 3\, s(n+3)^2\, s(n+1)^2 + 9\, s(n+3)\, s(n+2)^2\, s(n+1) = 0 \tag{7}$$

> *solve*(**(7)**[2], $s(n+4)$)

$$-\frac{s(n+3)\left(3\, s(n+3)\, s(n+1)^2 - 4\, s(n+3)\, s(n)\, s(n+2) + 9\, s(n+1)\, s(n+2)^2\right)}{s(n+3)\, s(n+1)\, s(n) + s(n)\, s(n+2)^2 - 6\, s(n+1)^2\, s(n+2)} \tag{8}$$

>

The linear algebra approach

```
>
> CPUTime(DalgSeq:-CCfiniteToSimpleRatrec(C2feq1, s(n), [Cfeq1, Cfeq2], [c[1](n),
      c[0](n)]))
```

$$0.187, s(n+4) = -\frac{s(n+3)\left(3\,s(n+3)\,s(n+1)^2 - 4\,s(n+3)\,s(n)\,s(n+2) + 9\,s(n+1)\,s(n+2)^2\right)}{s(n+3)\,s(n+1)\,s(n) + s(n)\,s(n+2)^2 - 6\,s(n+1)^2\,s(n+2)} \quad \textbf{(9)}$$

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